

Task 5 – Exploratory Data Analysis (EDA) Report

Dataset: Titanic Dataset (train.csv)

Objective

Perform Exploratory Data Analysis using Python, Pandas, Matplotlib and Seaborn to identify patterns, trends, relationships and anomalies in the Titanic dataset.

Analysis Performed

Dataset overview using `info()`, `describe()` and `value_counts()`; missing value analysis; histograms; boxplots; countplots; scatterplots; correlation heatmap; and pairplot visualizations.

Key Findings

1. Female passengers had higher survival rates.
2. First-class passengers were more likely to survive.
3. Fare showed a positive relationship with survival.
4. Most passengers were between 20 and 40 years old.
5. Fare distribution was positively skewed and contained outliers.
6. Cabin contained a large number of missing values.
7. Passenger class strongly influenced survival probability.

Conclusion

EDA successfully revealed important insights about passenger demographics, ticket fares, and survival patterns. The analysis demonstrates how visual and statistical exploration can support data-driven decision making.